Lecture 3 — part 1 (SVM & Boosting)

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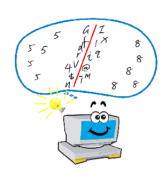
Advanced Data Sciences

SVM and Boosting



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Convexified ERM (ϕ-ERM)

Notation recall : ϕ-ERM

- Prediction function : $h : \mathcal{X} \to \mathbb{R}$
  $\hookrightarrow$ resulting classifier $g = \mathrm{sgn}(h)$
- Convex surrogate $\phi : \mathbb{R} \to \mathbb{R}$
- Training sample $\mathcal{D}_n = \{(X_i, Y_i),\ i = 1, \ldots, n\}$ ($\mathcal{Y} = \{-1, 1\}$)
- Risk and empirical ϕ-risk of h

$$A_P(h) = \mathbf{E}_P[\phi(-Yh(X))], \qquad \widehat{A}_n(h) = \frac{1}{n}\sum_{i=1}^{n} \phi(-Y_i h(X_i))$$

cf slide 12

$\phi_0(-Y_i h(x_i)) = \mathbb{1}(1 - Y_i h(x_i)) \geq 0$

$\alpha^T X_i$

- Main classifiers
  - Optimal prediction function for the ϕ-risk
    $$h_P^* = \arg\min_h A_P(h)$$
  - Optimal prediction function in $\mathcal{H} \subset \mathcal{F}(\mathcal{X}, \mathbb{R})$
    $$h_{P,\mathcal{H}}^* = \arg\min_{h \in \mathcal{H}} A_P(h)$$
  - Empirical minimizer of the ϕ-risk (ϕ-ERM)
    $$\widehat{h}_{n,\mathcal{H}} \in \arg\min_{h \in \mathcal{H}} \widehat{A}_n(g), \qquad \text{with } \mathcal{H} \subset \mathcal{F}(\mathcal{X}, \mathbb{R})$$

General bound based on Rademacher complexity

- Recall the Rademacher complexity of a family $\mathcal{H} \subset \mathcal{F}(\mathcal{X}, \mathbb{R})$

$$\mathcal{C}_n^{\mathrm{Rad}}(\mathcal{H}) = \sup_{\mathbf{x}_n = (x_1, \ldots, x_n) \in \mathcal{X}^n} \mathbf{E}_\sigma \left[\sup_{h \in \mathcal{H}} \left| \frac{1}{n} \sum_{i=1}^{n} \sigma_i h(x_i) \right| \right]$$

// Notion of Complexity

Theorem (Vapnik-Chervonenkis)

Let $0 < R < +\infty$. Assume ϕ is α- Lipschitz ($\alpha > 0$), convex surrogate on $[-R, R]$. Assume $\sup_{x \in \mathcal{X}} |h(x)| \leq R$. Then for all $\varepsilon > 0$, with probability greater that $1 - \varepsilon$

$$A_P(h_{n,\mathcal{H}}) - A_P(h_{P,\mathcal{H}}^*) \leq 8\alpha\, \mathcal{C}_n^{\mathrm{Rad}}(\mathcal{H}) + 2\Delta\phi(R)\sqrt{\frac{\ln((1/\varepsilon)}{n}}$$

where $\Delta\phi(R) = |\phi(R) - \phi(-R)|$

Corollary : VC-class

Theorem

If $\mathcal{H}$ (is binary) has VC-dimension $V_{\mathcal{H}}(n)$ on a sample of size n. Then

$$\mathcal{C}_n^{\mathrm{Rad}}(\mathcal{H}) \leq \sqrt{2V_{\mathcal{H}}(n)\frac{\ln(2(n+1))}{n}}$$

- Due to $\boxed{\mathbf{E}_\sigma[\sup_{\mathbf{t} \in \mathcal{T}} |\langle \boldsymbol{\sigma}, \mathbf{t} \rangle|] \leq \sqrt{2n \ln(2|\mathcal{T}|)}}$ [Maximal Inequality]
  $\hookrightarrow$ Set $\mathcal{T}$ is in bijection with $\mathrm{Tr}_{\mathcal{H}}(\{x_1, \ldots, x_n\})$
- Sauer Lemma : $|\ \mathrm{Tr}_{\mathcal{H}}(\{x_1, \ldots, x_n\})\ | \leq (n+1)^{V_{\mathcal{H}}(n)}$

For more general $\mathcal{H}$

- Shattering coefficient of $\mathcal{H}$ (cardinal of all possible labelings)

$$\mathbb{S}_n(\mathcal{H}) = \sup_{\mathbf{x}_n = (x_1, \ldots, x_n) \in \mathcal{X}^n} \{|(h(x_1), \ldots, h(x_n))|, \quad h \in \mathcal{H}\}$$

- If h is binary, $\{(h(x_1), \ldots, h(x_n)),\ h \in \mathcal{H}\}$ is in bijection with $\mathrm{Tr}_{\mathcal{H}}(\{x_1, \ldots, x_n\})$

Corollary : Covering numbers

- $\mathcal{H} \subset \mathcal{F}(\mathcal{X}, \mathbb{R})$ with metric d set
- Let $\varepsilon > 0$. An ε-net of $(\mathcal{H}, d)$ is a set V s.t.
  $$\forall h \in \mathcal{H},\ \exists h' \in V : \quad d(h, h') \leq \varepsilon$$
- The **Covering number** of $(\mathcal{H}, d)$ is
  $$\mathcal{N}(\mathcal{H}, d, \varepsilon) = \inf\{|V| : \quad V \text{ is a } \varepsilon\text{-net}\}$$

Theorem

If $h \in [-1, 1]$ for all $h \in \mathcal{H}$

$$\mathcal{C}_n^{\mathrm{Rad}}(\mathcal{H}) \leq \inf_{\varepsilon \geq 0} \left\{ \varepsilon + \sqrt{\frac{2\ln(2\mathcal{N}(\mathcal{H}, d_1^{\mathbf{x}_n}, \varepsilon))}{n}} \right\}$$

where $d_1^{\mathbf{x}_n}(h, h') = \frac{1}{n}\sum_{i=1}^{n} |h(x_i) - h'(x_i)|$

- Go further with *Chaining* ! ⇝ win log terms.
- Combine with fat-dimension to get a bound on the covering number in general settings

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Important particular cases

Again the notation

- Training sample $\mathcal{D}_n = \{(X_i, Y_i),\ i = 1, \ldots, n\}$ with $\mathcal{Y} = \{-1, 1\}$
- Convex "loss" function $\phi : \mathbb{R} \to \mathbb{R}_+$
- Class of functions $\mathcal{H} \subset \mathcal{F}(\mathcal{X}, \mathbb{R})$
- We define de ϕ-ERM

$$\widehat{h}_{n,\mathcal{H}} = \arg\min_{h \in \mathcal{H}} \underbrace{\frac{1}{n}\sum_{i=1}^{n} \phi(-Y_i h(X_i))}_{\text{empirical } \phi\text{-risk } \widehat{A}_n(h)}$$

- Classifier $g(\cdot) = \mathrm{sgn}\,(h(\cdot))$

Question : How should we choose ϕ and $\mathcal{H}$?
$\hookrightarrow$ Popular choices are well studied ! SVM and boosting

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Support Vector Machines (SVM)

Introduction to SVM

- Hilbert space $\mathcal{H}$ with scalar product $\langle \cdot, \cdot \rangle$
- Induced norm : $\|h\|_{\mathcal{H}} = \sqrt{\langle h, h \rangle}$
- For all $t > 0$, we define

$$\hat{h}_t \in \arg\min_{h \in \mathcal{H}: \|h\|_{\mathcal{H}} \leq t} \frac{1}{n}\sum_{i=1}^{n} \phi(-Y_i h(X_i)) \qquad (1)$$

Lemma

For all $t > 0$, there exists $\lambda > 0$ such that the solution of (1) is also solution of

$$\tilde{h}_\lambda \in \arg\min_{h \in \mathcal{H}} \frac{1}{n}\sum_{i=1}^{n} \phi(-Y_i h(X_i)) + \lambda\|h\|_{\mathcal{H}}^2.$$

⇝ regularization

And the converse is also true

$(1+u)_+$

Remarks :

- This is a convex optimization problem
- The classical choice for SVM is the Hinge loss $\phi(u) = (1+u)_+$
  but it works with any convex ϕ
- t is a tuning parameter that compromises estimation and approximation errors – cross-validation

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RKHS

Most important for SVM : $\boxed{\mathcal{H} \text{ is induced by a kernel } K}$

- $K : \mathcal{X} \times \mathcal{X} \to \mathbb{R}$ symmetric and positive-definite

  [symmetric] $\forall x, x' \in \mathcal{X}$ $K(x, x') = K(x', x)$
  [def.-pos.] $\forall x_1, \ldots, x_m \in \mathcal{X}$ the matrix $[K(x_j, x_{j'})]_{j,j'=1}^m$ is positive definite

- Functions space spanned by $x \mapsto K(x, a)$ for all $a \in \mathcal{X}$

$$\mathcal{H} = \left\{ h : \exists \alpha_1, \ldots, \alpha_m \in \mathbb{R},\ x_1, \ldots, x_m \in \mathcal{X} \text{ s.t. } h = \sum_{j=1}^{m} \alpha_j K(\cdot, x_j) \right\}$$

⇝ span of $K(\cdot, \cdot)$

Properties :

$$\langle h, h' \rangle = \sum_{j=1}^{m}\sum_{j'=1}^{m'} \alpha_j \alpha'_{j'} K(x_j, x'_{j'}), \qquad \text{with } h = \sum_{j=1}^{m} \alpha_j K(\cdot, x_j),\ h' = \sum_{j'=1}^{m'} \alpha'_{j'} K(\cdot, x'_{j'})$$

- We call $\mathcal{H}$ Reproducing Kernel Hilbert Space (RKHS)

Typical example. $\mathcal{X} = \mathbf{R}^d$ and $K(x, \tilde{x}) = \langle x, \tilde{x} \rangle = x^\top \tilde{x}$. Then

$\mathcal{H} = \left\{ h :\ \forall x \in \mathbf{R}^n,\ \exists x_h \in \mathbf{R}^n \text{ s.t. } h(x) = x_h^\top x \right\}$ linear function

$\|h\|_{\mathcal{H}}^2 = \|x_h\|_2^2$ $\|\cdot\|_2$ is the Euclidean norm

ex : $K(x, x') = e^{-\frac{\|x - x'\|^2}{2\sigma^2}}$
Gaussian kernel.

Efficient solution for SVM

$\mathcal{D}_n = ((X_i, Y_i)\ i = 1 \ldots n)$ Training set.

Theorem

For some RKHS $\mathcal{H}$ and $\lambda > 0$, the solution

$$\tilde{h}^{\mathrm{SVM}} \in \arg\min_{h \in \mathcal{H}} \left(\frac{1}{n}\sum_{i=1}^{n} \phi(-Y_i h(X_i)) + \lambda\|h\|_{\mathcal{H}}^2 \right)$$

is given by

$$\tilde{h}^{\mathrm{SVM}}(\cdot) = \sum_{j=1}^{n} \alpha_j^{\mathrm{SVM}} K(\cdot, X_j)$$

WAW!

where

$$\boldsymbol{\alpha}^{\mathrm{SVM}} \in \arg\min_{\boldsymbol{\alpha} \in \mathbb{R}^n} \left\{ \frac{1}{n}\sum_{i=1}^{n} \phi(-Y_i \boldsymbol{\alpha}^\top \mathbf{K}\mathbf{e}_i) + \lambda\, \boldsymbol{\alpha}^\top \mathbf{K} \boldsymbol{\alpha} \right\}$$

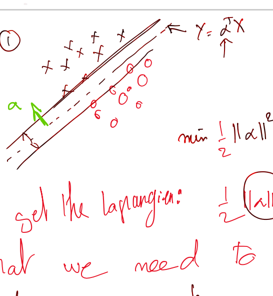
with $\mathbf{K}_{i,j} = (K(X_i, X_j))_{i,j} \in \mathbf{R}^{n \times n}$ and $\{\mathbf{e}_j\}_j$ is the canonical basis in $\mathbf{R}^n$

Remarks

- $\boldsymbol{\alpha}^{\mathrm{SVM}}$ is solution of an optimization problem in dimension n
- For solutions $\tilde{\alpha} \neq \tilde{\alpha}'$, we have $\sum_{i=1}^{n} \tilde{\alpha}_i K(\cdot, X_i) = \sum_{i=1}^{n} \tilde{\alpha}'_i K(\cdot, X_i)$
- SVM with hinge or quadratic loss can be solved by SOCP

Fig ①

normal vector to $y = \alpha^T x$



$\max M \Leftarrow \frac{1}{\|\alpha\|}$

st $Y_i(\alpha^T X_i) \geq 1$

$\min \frac{1}{2}\|\alpha\|^2$ st : $Y_i(\alpha^T X_i) \geq 1$

We get the Lagrangian: $\frac{1}{2}\|\alpha\|^2 + \sum_i c_i (Y_i \alpha^T X_i) - 1)$

That we need to minimize in α

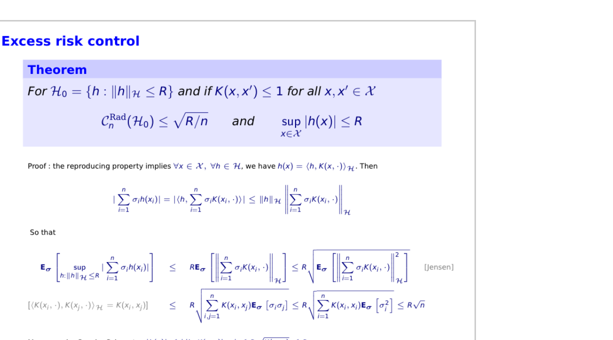
After some work ... (introducing slack variables and dealing with dual problem) we get

$\min_\alpha \frac{1}{2}\|\alpha\|^2 + \lambda \sum \phi(-Y_i \alpha^T X_i)$ where $\phi = (1+u)_+$ Hinge Loss

What is important is that the solution depends only on $\langle X_i, X_j \rangle$ and $h(x)$ depends on $\langle X_i, x \rangle$ and $\langle X_i, X_j \rangle$

Then kernel Trick : we can replace $\langle \cdot, \cdot \rangle \xrightarrow{\text{by}} \langle \cdot, \cdot \rangle_{\mathcal{H}}$

②



Qu°: How can we find φ ? We do not care!

What matters is just the kernel.

In applications, we just try several kernels and choose a nice one ⇝ popul kernels are linear, gaussian, exponential, ...

Excess risk control

Theorem

For $\mathcal{H}_0 = \{h : \|h\|_{\mathcal{H}} \leq R\}$ and if $K(x, x') \leq 1$ for all $x, x' \in \mathcal{X}$

$$\mathcal{C}_n^{\mathrm{Rad}}(\mathcal{H}_0) \leq \sqrt{R/n} \qquad \text{and} \qquad \sup_{x \in \mathcal{X}} |h(x)| \leq R$$

Proof : the reproducing property implies $\forall x \in \mathcal{X},\ \forall h \in \mathcal{H}$, we have $h(x) = \langle h, K(x, \cdot) \rangle_{\mathcal{H}}$. Then

$$|\sum_{i=1}^{n} \sigma_i h(x_i)| = |\langle h, \sum_{i=1}^{n} \sigma_i K(x_i, \cdot) \rangle| \leq \|h\|_{\mathcal{H}} \left\| \sum_{i=1}^{n} \sigma_i K(x_i, \cdot) \right\|_{\mathcal{H}}$$

So that

$$\mathbf{E}_\sigma \left[\sup_{h : \|h\|_{\mathcal{H}} \leq R} |\sum_{i=1}^{n} \sigma_i h(x_i)| \right] \leq R\mathbf{E}_\sigma \left[\left\| \sum_{i=1}^{n} \sigma_i K(x_i, \cdot) \right\|_{\mathcal{H}} \right] \leq R\sqrt{\mathbf{E}_\sigma \left[\left\| \sum_{i=1}^{n} \sigma_i K(x_i, \cdot) \right\|_{\mathcal{H}}^2 \right]} \quad \text{[Jensen]}$$

$$[\langle K(x_i, \cdot), K(x_j, \cdot) \rangle_{\mathcal{H}} = K(x_i, x_j)] \quad \leq R\sqrt{\sum_{i,j=1}^{n} K(x_i, x_j)\mathbf{E}_\sigma[\sigma_i \sigma_j]} \leq R\sqrt{\sum_{i=1}^{n} K(x_i, x_i)\mathbf{E}_\sigma[\sigma_i^2]} \leq R\sqrt{n}$$

Moreover, by Cauchy Schwartz : $|h(x)| \leq |\langle h, K(x, \cdot) \rangle_{\mathcal{H}}| \leq R\sqrt{K(x, x)} \leq R$

Boosting

Optimization problem for boosting

- Let $\mathcal{G} \subset \mathcal{F}(\mathcal{X}, \{\pm 1\})$ be a family of simple classifiers
  $\hookrightarrow$ For instance *linear* or *piecewise constant* functions
- λ-envelop of $\mathcal{G}$

$$\mathcal{H}_\lambda = \left\{ \sum_{j=1}^{m} \alpha_j g_j : m \in \mathbb{N},\ \begin{matrix} \boldsymbol{\alpha} \in \mathbb{R}_+^m, \\ (g_1, \ldots, g_m) \in \mathcal{G}^m \end{matrix},\ \sum_{j=1}^{m} \alpha_j \leq \lambda \right\}$$

- **Remark :** $\mathcal{H}_\lambda$ is convex for any $\lambda > 0$ *(Exercise : Show it !)*
- Boosting prediction function is given by

$$\hat{h}_\lambda^{\mathrm{boost}} \in \arg\min_{h \in \mathcal{H}_\lambda} \hat{A}_n(h) = \arg\min_{h \in \mathcal{H}_\lambda} \frac{1}{n}\sum_{i=1}^{n} \phi(-Y_i h(X_i)).$$

- Corresponding classifier : $\hat{g}_\lambda^{\mathrm{boost}} = \mathrm{sgn}\left(\hat{h}_\lambda^{\mathrm{boost}}\right)$
- Tuning parameter λ – Cross-validation

Consistency of Boosting

Theorem (Lugosi and Vayatis (2004))

If $\lambda = \lambda_n \to \infty$ is s.t. $\lambda_n \phi'(\lambda_n)\sqrt{\frac{\ln n}{n}} \to \infty$ and $\mathcal{G}$ is s.t.

a) $\lim_{\lambda \to \infty} \inf_{h \in \mathcal{H}_\lambda} A(h) = \inf_{h \in \mathcal{F}(\mathcal{X}, \mathbb{R})} A(h)$ *[$\mathcal{H}_\lambda$ contains the optimal]*

b) *$\mathcal{G}$ is a VC-class*

Then $\hat{h}_\lambda^{\mathrm{boost}}$ is universally consistent

- Boosting optimization problem is convex but hard to solve since $\mathcal{H}_\lambda$ is infinite dimensional
- The "real" Boosting is with $\phi(u) = e^u$ but any convex function works
- **AdaBoost** : iterative solver to approximate $\hat{h}_\lambda^{\mathrm{boost}}$
  $\hookrightarrow$ No theoretical guarantees.

Excess risk control

Theorem

For $\mathcal{H}_R = \left\{ h : \sum_{j=1}^{m} \alpha_j g_j : \sum_{j=1}^{m} |\alpha_j| \leq R \right\}$

$$\mathcal{C}_n^{\mathrm{Rad}}(\mathcal{H}_R) \leq R\sqrt{\frac{2\ln(2m)}{n}}$$

Proof : The mapping $x \mapsto \sum_{i=1}^{n} \sigma_i \sum_{j=1}^{m} \alpha_j g_j(x)$ is linear. Then it reaches its maximum and minimum over the ℓ_1 ball $\mathcal{B}_1$ at one of the vertices of $\mathcal{B}_1$. So that

$$\mathbf{E}_\sigma \left[\sup_{h \in \mathcal{H}_R} |\sum_{i=1}^{n} \sigma_i h(x_i)| \right] = R\mathbf{E}_\sigma \left[\max_{j \in [m]} |\sum_{i=1}^{n} \sigma_i g_j(x_i)| \right]$$
$$\leq R\sqrt{2n\ln(2m)} \qquad \text{[Maximal Inequality]}$$

$\max_{\alpha_1 \ldots \alpha_m} \sum_{j=1}^{m} \alpha_j g_j \to \max_j R g_j$